

Electrical design Philosophy:

1. Power Architecture & Battery Selection

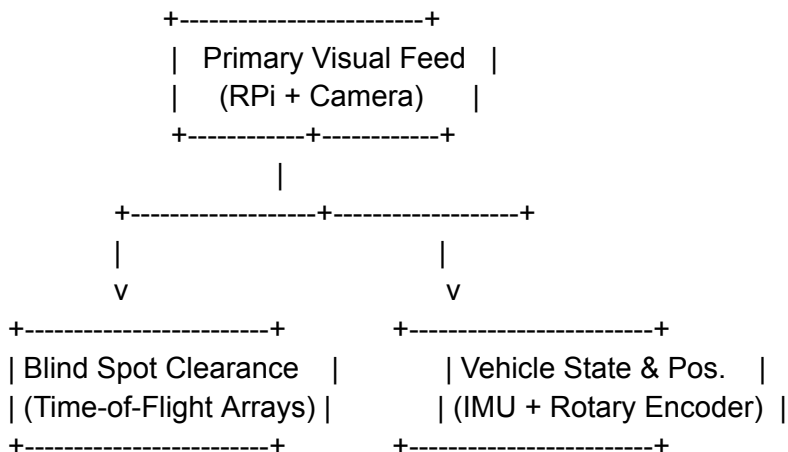
- **Cell Chemistry Selection (LiPo):** We selected a Lithium Polymer (LiPo) battery specifically to optimize the vehicle's power-to-weight ratio. LiPo technology provides a high discharge rate (C-rating) necessary to handle rapid motor acceleration without voltage drops, while maintaining a compact physical volume critical for tight chassis constraints.
- **Thermal & Electrical Efficiency:** To supply power to our logic and sensing units, we utilized high-efficiency switching **Buck Converters** instead of linear regulators (such as LM7805). This significantly reduces thermal dissipation and avoids wasting energy as heat, extending battery operational life during testing and runs.
- **Discrete Motor Driver Architecture:** We opted for a discrete MOSFET H-Bridge driver rather than an integrated shield. This choice drastically reduced the PCB footprint and simplified wire routing. The inherent body diodes inside the discrete MOSFETs handle inductive kickback, eliminating the need for external flyback diodes.

2. Noise Isolation & Topology

- **Inherent Power Isolation:** Our architectural layout naturally separates clean logic lines from high-noise inductive loads. High-noise components (motors and power drivers) draw directly from the LiPo battery or dedicated buck converter lines.
- **Signal Protection:** With high-current paths isolated from the central logic, low-voltage control signals—specifically PWM motor drive commands and high-frequency rotary encoder pulses—remain protected from electromagnetic interference (EMI) and ground bounces.

3. Sensor Fusion & Odometry Strategy

To achieve robust autonomous navigation on the track, we implemented a complementary sensor suite designed to eliminate visual blind spots and maintain high-precision position tracking:



- **Visual Perception + Proximity Sensing:** While the primary camera provides long-range track layout recognition, it suffers from localized blind spots directly around the vehicle bumpers. We paired the camera with **Time-of-Flight (ToF)** distance sensors to cover these blind spots and provide accurate distance-to-wall measurements.
- **Odometry & Dead Reckoning:** To compensate for momentary visual occlusion or lighting variations, an **IMU** (heading/angular velocity) and **rotary encoders** (wheel rotation/linear distance) run continuously to provide steady, multi-sensor vehicle state estimation.

4. Communication & Serviceability

- **Processing Bus (UART):** For inter-board communication between the main vision processor (Raspberry Pi) and the low-level real-time controller (Arduino Nano), we selected **UART**. UART provides a lightweight, point-to-point serial connection that avoids the master-slave overhead, complex addressing, and driver complexity of I2C or SPI buses.
- **Modular Cabling (JST Connectors):** To ensure high serviceability in pit conditions, all sensors, power rails, and actuators use polarized **JST connectors**. This modularity allows instant component swapping during troubleshooting without requiring re-soldering.

5. Easy compute bus access and wiring

- All of our compute components (i.e raspberry pi zero w and arduino nano) as well as additional components like the multiplexer, IMU, and gyroscope are placed at the top of the vehicle in a detachable manner to allow ease of debugging, wiring, access, last minute changes, flashing code, etc. Wiring and components are placed on a perf board